COMPUTER NETWORKS REVISION GUIDE

A Protocol-Driven Reference on Network Layers, Transport Reliability, Routing & Modern Web Protocols

A Concise, Visual, and Mechanism-Focused Interview & Revision Handbook

METADATA FIELD	SPECIFICATION
Subject:	Computer Networks & Distributed Communication
Target Audience:	SDE / Technical Interviews (Top Tech, FAANG, Product Firms)
Methodology:	Internal Mechanics • State Transitions • Verified Code Recipes • Trade-offs
Source File Consolidated:	Interview Concepts/CN.py (659 lines covering 7 comprehensive parts)
Subject Coverage:	OSI 7 Layers, TCP/IP, IP Subnetting, Routing/Switching, TCP Handshake, Flow/Congestion Control, TLS, WebSockets
Key Architectural Focus:	Packet Encapsulation, Sequence/ACK Numbers, Congestion Math, State Machines & Protocol Sequences
Target Evaluation:	Technical Interview Placement Preparation - Core Engineering Track

CH	CHAPTER TITLE & CORE MECHANISM	CRITICAL INTERVIEW CONCEPTS & PROTOCOLS
1	Network Architectures: OSI 7-Layer vs TCP/IP	7 Layers breakdown, Duties, Service Boundaries, PDU Types
2	Data Encapsulation & PDU Hierarchy	Data -> Segment -> Packet -> Frame -> Bits; Header decapsulation
3	IP Addressing, CIDR Subnetting & ARP	IPv4 vs IPv6, Private IP Ranges, Subnet Masks, ARP vs NDP
4	Network Layer Routing vs L2 Switching	MAC table forwarding vs IP Routing, Link-State OSPF vs Distance-Vector RIP
5	TCP Architecture & Connection Lifecycle	3-Way Handshake (SYN/ACK), Sequence numbers, 4-Way Teardown, TIME_WAIT
6	TCP Flow Control & Congestion Control	Sliding Window, rwnd, Slow Start, Congestion Avoidance, AIMD math
7	UDP Protocol & Comparison with TCP	Datagram structure, Unreliable delivery, Video/VoIP performance profile
8	Modern Transport: QUIC & HTTP/3	UDP-based transport, 0-RTT connection, No Head-of-Line blocking
9	DNS, HTTP/1.1, HTTP/2, HTTPS & TLS Handshake	DNS recursion, HTTP/1.1 vs HTTP/2 multiplexing, TLS 1.3 Asymmetric/Symmetric keys
10	Real-Time Web: WebSockets vs SSE vs Polling	HTTP Upgrade Handshake, Full-Duplex binary framing, SSE, Long Polling

CHAPTER 1 - NETWORK ARCHITECTURES: OSI 7-LAYER VS TCP/IP

Network architectures decompose complex distributed communications into modular functional layers. Each layer serves the layer above it and utilizes services provided by the layer below it through well-defined protocol boundaries.

1.1 The OSI 7-Layer Reference Model

Layer # & Name	Protocol Data Unit	Primary Responsibility	Key Protocols & Hardware
7. Application	Data (Payload)	User interface, network application services.	HTTP, HTTPS, DNS, SMTP, FTP, WebSockets, SSH
6. Presentation	Data	Data translation, syntax formatting, encryption (TLS), compression.	SSL/TLS, JPEG, ASCII, JSON, Protobuf
5. Session	Data	Authentication, session establishment, checkpointing, termination.	RPC, NetBIOS, PPTP
4. Transport	Segment (TCP) / Datagram (UDP)	End-to-end process-to-process delivery, port multiplexing, reliability.	TCP, UDP, QUIC; Ports (0..65535)
3. Network	Packet	Logical host-to-host addressing, packet routing across networks.	IP (IPv4, IPv6), ICMP, OSPF, BGP; Routers
2. Data Link	Frame	Physical hop-to-hop transfer, MAC addressing, local framing, error detection.	Ethernet (802.3), Wi-Fi (802.11), ARP, VLAN; Switches
1. Physical	Bits	Transmission of raw electrical, optical, or radio signals over media.	Cables (Cat6, Fiber), Hubs, Modems, Repeaters

1.2 TCP/IP 4-Layer Architecture Mapping

While OSI is the theoretical conceptual model, the Internet runs on the pragmatic **TCP/IP 4-Layer Protocol Suite**:

- **Application Layer (OSI 5, 6, 7):** Handles all high-level application protocols (HTTP, DNS, SSH).
- **Transport Layer (OSI 4):** Manages host-to-host process delivery (TCP, UDP).
- **Internet Layer (OSI 3):** Routes packets across distinct networks (IP, ICMP).
- **Network Access / Link Layer (OSI 1, 2):** Handles local hardware framing and physical signaling (Ethernet, Wi-Fi).

CHAPTER 2 - DATA ENCAPSULATION & PDU HIERARCHY

When an application sends data across the Internet, the payload descends through the protocol stack. At each layer, a protocol header (and sometimes trailer) is prepended—a process called **Encapsulation**.

2.1 Encapsulation Journey (Client to Server)

1. **Application Layer:** User issues HTTP GET request -> Data payload.
2. **Transport Layer:** Prepends TCP Header (Source Port, Destination Port, Sequence Number, ACK Number) -> **TCP Segment**.
3. **Network Layer:** Prepends IP Header (Source IP, Destination IP, TTL, Protocol ID) -> **IP Packet**.
4. **Data Link Layer:** Prepends Ethernet Header (Source MAC, Next-Hop MAC) and appends CRC Trailer -> **Ethernet Frame**.
5. **Physical Layer:** Encodes frame bytes into electrical voltage transitions or photon pulses -> **Bits on wire**.
6. **Hop-by-Hop Decapsulation:** Routers inspect and strip Layer 2 Frame headers to read the Layer 3 Destination IP, determine the next hop, update the TTL, and re-encapsulate into a new Layer 2 Frame with the next router's MAC address!

2.2 Network Boundary Units Comparison

PDU Term	Operating Layer	Addressing Used	Primary Function
Segment / Datagram	Layer 4 (Transport)	Port Numbers (e.g., 443, 80)	Multiplexes data to the specific destination process.
Packet	Layer 3 (Network)	IP Addresses (e.g., 142.250.190.46)	Routes data across global internetworks.
Frame	Layer 2 (Data Link)	MAC Addresses (e.g., 00:1A:2B:3C:4D:5E)	Delivers bits between directly connected local network interfaces.

CHAPTER 3 - IP ADDRESSING, CIDR SUBNETTING & ARP

3.1 IPv4 vs IPv6 Architectural Comparison

Feature	IPv4	IPv6
Address Length	32 bits (4 octets: 192.168.1.1)	128 bits (8 hex hextets: 2001:0db8::8a2e:0370:7334)
Total Address Space	2^32 ≈ 4.29 Billion addresses (exhausted!)	2^128 ≈ 3.4 x 10^38 addresses (effectively infinite)
Header Size	Variable (20 to 60 bytes with options)	Fixed (40 bytes; simplifies router processing)
Checksum	Yes (calculated at every hop; high router overhead)	None (relies on L2 CRC and L4 transport checksum)
Address Resolution	ARP (broadcast-based)	NDP (Neighbor Discovery Protocol, multicast-based)
IP Configuration	Manual or DHCP	Stateless Address Autoconfiguration (SLAAC) or DHCPv6

3.2 Private IP Ranges & CIDR Subnetting

RFC 1918 defines non-routable private addresses for internal local networks (translated to public IPs via NAT):
• **Class A:** 10.0.0.0/8 (10.0.0.0 - 10.255.255.255 -> 16.7M hosts)
• **Class B:** 172.16.0.0/12 (172.16.0.0 - 172.31.255.255 -> 1M hosts)
• **Class C:** 192.168.0.0/16 (192.168.0.0 - 192.168.255.255 -> 65,536 hosts; home Wi-Fi)
• **Subnet Calculation Rule:** A /24 subnet mask has 24 network bits and 32 - 24 = 8 host bits. Total addresses = 2^8 = 256. Usable host addresses = 256 - 2 = 254 (subtracting Network Address .0 and Broadcast Address .255).

3.3 ARP (Address Resolution Protocol)

When Host A wants to send a frame to Host B on the same local subnet, it knows Host B's IP but lacks its physical MAC address.
1. Host A broadcasts an **ARP Request** (Destination MAC: FF:FF:FF:FF:FF:FF): *'Who has IP 192.168.1.50? Tell 192.168.1.10'*.
2. All local nodes receive the broadcast; only Host B matches the IP.
3. Host B unicasts an **ARP Reply** with its MAC address.
4. Host A caches the mapping in its local ARP table for future frames.

CHAPTER 4 - NETWORK LAYER ROUTING VS L2 SWITCHING

4.1 Switch (Layer 2) vs Router (Layer 3)

Dimension	Layer 2 Switch	Layer 3 Router
Operating Layer	Data Link Layer (L2)	Network Layer (L3)
Forwarding Table	MAC Address Table (learned via source MAC of frames)	Routing Table (configured via static routes or BGP/OSPF)
Scope of Operation	Connects devices inside the **same local subnet/LAN**.	Connects **different independent subnets / WANs**.
Broadcast Domain	Single broadcast domain (broadcasts flood all ports).	Breaks broadcast domains (does NOT forward broadcasts).
Collision Domain	Each switch port is an isolated collision domain.	Each router interface is an isolated collision domain.

4.2 Routing Algorithms: Link-State vs Distance-Vector

Algorithm Type	Core Graph Algorithm	Routing Protocol	Mechanism & Convergence
Link-State	Dijkstra's Shortest Path First	OSPF (Open Shortest Path First)	Every router floods link-state advertisements (LSAs) to build full topology map. Fast convergence; high CPU/memory.
Distance-Vector	Bellman-Ford Algorithm	RIP (Routing Information Protocol)	Routers share periodic distance tables with immediate neighbors only. Slow convergence; suffers from Count-to-Infinity problem.

CHAPTER 5 - TCP ARCHITECTURE & CONNECTION LIFECYCLE

Transmission Control Protocol (TCP) is a connection-oriented, reliable, byte-stream protocol. It guarantees ordered, error-checked delivery through sequence numbers, acknowledgments, and retransmissions.

5.1 The TCP 3-Way Handshake (Connection Establishment)

Before data can transfer, client and server synchronize their Initial Sequence Numbers (ISNs):
1. **Step 1 (SYN):** Client -> Server. Flags: SYN = 1. Sequence Number = x. (Client enters SYN_SENT).
2. **Step 2 (SYN-ACK):** Server -> Client. Flags: SYN = 1, ACK = 1. Sequence Number = y, Acknowledgment Number = x + 1. (Server enters SYN_RCVD).
3. **Step 3 (ACK):** Client -> Server. Flags: ACK = 1. Sequence Number = x + 1, Acknowledgment Number = y + 1. (Both enter ESTABLISHED). Data payload may accompany this step.

5.2 The TCP 4-Way Handshake (Connection Teardown)

Because TCP is full-duplex, each direction of communication must close independently:
1. **Step 1 (FIN):** Client -> Server. *'I have no more data to send'*. (Client enters FIN_WAIT_1).
2. **Step 2 (ACK):** Server -> Client. Acknowledges client's FIN. (Server enters CLOSE_WAIT; client enters FIN_WAIT_2). Server can still transmit pending data!
3. **Step 3 (FIN):** Server -> Client. Server finishes data transmission and sends its FIN. (Server enters LAST_ACK).
4. **Step 4 (ACK):** Client -> Server. Client acknowledges server's FIN and enters **TIME_WAIT** for 2 * MSL (Maximum Segment Lifetime, typically 60-120s).

WHY IS TIME_WAIT STATE MANDATORY?
1. **Guarantees Final ACK Delivery:** If Client's final ACK is lost, Server re-sends its FIN. Client must remain alive in TIME_WAIT to re-ACK; otherwise Server gets a connection reset (RST).
2. **Drains Stray Segments:** Ensures delayed, duplicated packets from the old connection die out in the internet before a new connection re-uses the same port tuple.

CHAPTER 6 - TCP FLOW CONTROL & CONGESTION CONTROL

6.1 Flow Control (Receiver Protection via Sliding Window)

Problem: Fast sender overwrites slow receiver's buffer.
Mechanism: Receiver advertises its available buffer space in the TCP Header via **Receiver Window (rwnd)**. The sender guarantees: LastByteSent - LastByteAcked <= rwnd. If rwnd == 0, sender stops transmitting and periodically sends 1-byte *Window Probes* to detect when buffer space frees.

6.2 Congestion Control (Network Protection via AIMD)

Problem: Excessive packets saturate intermediate router buffers, causing massive packet loss and collapse.
Sender maintains a dynamic **Congestion Window (cwnd)**. Total unacknowledged data cannot exceed min(cwnd, rwnd).

Phase	Condition / Trigger	Window Growth Math	Internal Objective
1. Slow Start	cwnd < Slow Start Threshold (ssthresh)	**Exponential Growth:** cwnd doubles every RTT (cwnd += 1 per ACK).	Rapidly probe available network capacity from cold start.
2. Congestion Avoidance	cwnd >= ssthresh	**Linear Growth (Additive Increase):** cwnd += 1 per RTT.	Safely probe capacity near link saturation point.
3. Packet Loss: 3 Duplicate ACKs	Fast Retransmit triggered	**Multiplicative Decrease:** ssthresh = cwnd / 2, cwnd = ssthresh (Fast Recovery).	Mild packet loss; avoids dropping cwnd back to 1.
4. Packet Loss: Timeout (RTO)	Retransmission Timer expires	**Hard Reset:** ssthresh = cwnd / 2, **cwnd reset to 1 MSS!** Slow start restarts.	Severe network congestion; back off completely.

CHAPTER 7 - UDP PROTOCOL & COMPARISON WITH TCP

7.1 UDP Architecture & Header Simplicity

User Datagram Protocol (UDP) is a minimal, lightweight, connectionless transport protocol. It provides zero reliability, no ordering, and no congestion control, acting only as a thin multiplexing wrapper over IP.

Dimension	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Model	Connection-oriented (Requires 3-way handshake).	Connectionless (Transmits immediately with zero setup latency).
Reliability	Guaranteed delivery via sequence numbers, ACKs & retransmission.	Best-effort delivery; packets can be lost, duplicated, or reordered.
Data Stream Format	Continuous byte stream (no application message boundaries).	Discrete message datagrams (preserves packet boundaries).
Header Overhead	Heavy: 20 bytes minimum (up to 60 bytes with options).	Lightweight: exactly 8 bytes fixed header.
Flow & Congestion	Built-in Flow Control (rwnd) and Congestion Control (AIMD).	None; sends at raw application speed.
Ideal Use Cases	Web pages (HTTP/HTTPS), file transfers (FTP), database queries, SSH.	Live video streaming, VoIP, online gaming, DNS queries, DHCP, QUIC.

CHAPTER 8 - MODERN TRANSPORT: QUIC & HTTP/3

8.1 Why TCP Is Replaced: Head-of-Line Blocking

In HTTP/2, multiple application streams are multiplexed over a single TCP connection. However, because TCP enforces strict byte-stream ordering, if a single TCP packet is dropped, **ALL multiplexed streams stall** until the missing packet is retransmitted—a critical flaw known as **TCP Head-of-Line (HoL) Blocking**.

8.2 QUIC Architectural Innovations (RFC 9000)

Feature	Standard TCP + TLS	QUIC Transport Protocol (HTTP/3)
Underlying Transport	Runs directly on IP (OS kernel space).	Built on top of UDP in user-space.
Connection Latency	1 RTT for TCP + 1 RTT for TLS 1.3 = 2 RTTs.	**0-RTT / 1-RTT** combined transport + TLS handshake.
Multiplexing & HoL	Single dropped packet stalls all streams.	Streams are independent! Dropping a packet in Stream A never stalls Stream B.
Connection Migration	Bound to 4-tuple (Src IP, Src Port, Dst IP, Dst Port). Switching Wi-Fi to 5G breaks TCP.	Uses 64-bit **Connection ID**. Mobile network switches survive seamlessly!

CHAPTER 9 - DNS, HTTP EVOLUTION & TLS HANDSHAKE

9.1 DNS Resolution Hierarchy: Step-by-Step

When resolving api.google.com:
1. Browser cache -> OS local cache -> Local DNS Resolver (ISP / 8.8.8.8).
2. Recursive Resolver queries **Root Nameserver (.)** -> returns TLD server IP for .com.
3. Resolver queries **TLD Nameserver (.com)** -> returns Authoritative nameserver IP for google.com.
4. Resolver queries **Authoritative Nameserver** -> returns final IP address (A record for IPv4, AAAA for IPv6).

9.2 HTTPS & TLS 1.3 Handshake Sequence

HTTPS encrypts HTTP traffic using Transport Layer Security (TLS). It uses **Asymmetric Encryption** (RSA / ECDHE) to authenticate identity and safely agree on a shared secret, then switches to **Symmetric Encryption** (AES-GCM / ChaCha20) for high-speed data encryption.

TLS 1.3 HANDSHAKE (1 RTT)

1. **ClientHello:** Supported cipher suites + Client Random + Client Key Share (Diffie-Hellman public key).
2. **ServerHello:** Selected cipher + Server Random + Server Key Share + Digital Certificate (signed by CA).
3. **Key Derivation:** Both parties independently derive the exact same Master Secret using ECDHE math.
4. **Encrypted Session:** All subsequent HTTP requests and responses travel fully encrypted.

CHAPTER 10 - REAL-TIME WEB: WEBSOCKETS VS SSE VS POLLING

Technology	Communication Direction	Connection Model	Protocol & Overhead	Best Use Case
Short Polling	Unidirectional (Client -> Server)	New HTTP connection every X seconds.	Standard HTTP; massive header overhead on every request.	Low-frequency dashboard checks.
Long Polling	Unidirectional (Client -> Server)	HTTP request hangs until server has data.	HTTP overhead per event; connection teardown on every message.	Legacy chat systems.
Server-Sent Events (SSE)	Unidirectional (**Server -> Client**)	Persistent, single HTTP connection.	Lightweight text/event-stream; automatic browser reconnection.	Stock tickers, live notifications, LLM token streaming.
WebSockets	**Bidirectional (Full-Duplex)**	Persistent TCP connection initiated via HTTP Upgrade.	Ultra-low framing overhead (2-10 bytes per frame); binary or text.	Multiplayer gaming, collaborative editing, interactive trading.

COMPUTER NETWORKS INTERVIEW CHEAT-SHEET

- **OSI vs TCP/IP:** OSI is 7 layers; TCP/IP is 4 layers.
- **TCP 3-Way Handshake:** SYN -> SYN-ACK -> ACK.
- **TCP Reliability:** Guaranteed via Sequence numbers, ACKs, Retransmit timers & Checksums.
- **Congestion Control:** Slow Start (exponential) -> Congestion Avoidance (AIMD linear) -> Fast Recovery.
- **HTTP Evolution:** HTTP/1.1 (sequential pipeline) -> HTTP/2 (multiplexed TCP) -> HTTP/3 (QUIC on UDP without HoL).